

Performance Evaluation of Federated Learning Algorithms

under IID and Non-IID Data Distributions

FedAvg

FedProx

SCAFFOLD

FedNova

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01

Introduction to Federated Learning

What FL is, why it matters, and how it works

02

IID vs. Non-IID Data Distributions

Statistical heterogeneity and the client drift problem

03

Research Objectives

What we aim to evaluate and measure

04

Algorithms: FedAvg · FedProx · SCAFFOLD · FedNova

How each algorithm addresses Non-IID challenges

05

System Design & Experimental Setup

Datasets, models, partitioning, configuration

06

Results & Analysis (4 Experiments)

Accuracy, loss, Dirichlet sweep, hyperparameter sensitivity

07

Statistical Validation

Welch's t-test, Bonferroni correction, Cohen's d

08

Key Findings & Conclusions

What we discovered — the Non-IID accuracy barrier

09

Future Work

Larger federations, new heterogeneity types, CUDA scaling

What is Federated Learning?

Definition

Federated Learning is a distributed machine learning model where multiple clients collaboratively train a shared global model without ever sharing their raw private data with a central server.

Each client trains locally on its own private dataset and only sends model parameters (weights) back to the server for aggregation.

Why It Matters



Privacy

Raw data never leaves the device



Efficiency

Reduced bandwidth consumption



Scalability

Works with millions of edge devices



Applications

Healthcare, Finance, IoT, Mobile

How Federated Learning Works

 **FEDERATED SERVER**
Global Model $w(t)$

① *Broadcast global model*

Client0
Classes {0,1}
Local SGD

Client1
Classes {2,3}
Local SGD

Client2
Classes {4,5}
Local SGD

Client3
Classes {6,7}
Local SGD

② *Local training (no data leaves!)*

③ **Aggregation:** $w(t+1) = \sum (n_k/n) \cdot w_k$

 Only model weights uploaded — ZERO raw data transmitted

↻ Repeat for $T = 50$ communication rounds

IID vs. Non-IID Data Distributions

✓ IID — Independent & Identically Distributed

What it means:

Each client receives a balanced random sample of the full dataset. Every client sees approximately equal proportions of all classes. Local optimization objectives closely match the global objective.

Effect on training:

- Local gradients are unbiased estimates of global gradient
- Fast, stable convergence to high accuracy
- All algorithms behave similarly and effectively

EMD = 0.059 (near-uniform distribution)

MNIST: 98.35% · CIFAR-10: 79.81%

⚠ Non-IID — Label-Skew Heterogeneity

What it means:

Each client receives data from only 2 of the 10 classes (class exclusion). Clients have completely different data distributions. Local and global objectives diverge sharply.

Effect on training:

- Local gradients are BIASED → client drift phenomenon
- Optimization diverges toward local client optima
- A structural accuracy barrier appears at ~65-66%

EMD = 1.938 (severe class exclusion)

MNIST: 65.69% · CIFAR-10: 27–34%

Each algorithm targets a different failure mode of naive weighted averaging under statistical heterogeneity.

FedAvg

McMahan et al., 2017

Baseline — Weighted Parameter Averaging

$$w(t+1) = \sum (n_k/n) \cdot w_k(t+1)$$

Clients perform local SGD for E epochs, then the server takes a weighted average of all local model parameters (weighted by dataset size).

✓ Simple, scalable, communication-efficient ✗ Convergence degrades sharply under Non-IID

FedProx

Li et al., 2020

Proximal Regularization to Limit Client Drift

$$\min F_k(w) + (\mu/2) \|w - w(t)\|^2$$

Adds a proximal term to each client's objective function, penalizing large deviations from the current global model. μ controls regularization strength.

✓ More stable under heterogeneous data ✗ Performance sensitive to μ selection

SCAFFOLD

Karimireddy et al., 2020

Variance Reduction via Control Variates

$$g_k(w) = \nabla F_k(w) - c_k + c$$

Each client maintains a local control variate c_k to debias its gradient estimate, correcting for the systematic gradient bias caused by data heterogeneity.

✓ Strong theoretical convergence guarantees ✗ Additional communication & memory overhead

FedNova

Wang et al., 2020

Normalized Aggregation for Objective Consistency

$$\Delta_k = (w(t) - w_k(t+1)) / \tau_k$$

Normalizes each client's update by its number of local SGD steps τ_k , then applies an effective step-size correction during aggregation to ensure consistent optimization.

✓ Corrects bias from unequal local steps ✗ More complex implementation

Why this research is needed:

Previous works only compared 1–2 algorithms, used simplistic heterogeneity assumptions, lacked statistical significance analysis, or evaluated on a single dataset. This study fills that gap with a comprehensive, reproducible benchmark.

01

Algorithm Comparison

Systematically evaluate FedAvg, FedProx, SCAFFOLD, and FedNova under identical experimental conditions across multiple heterogeneous environments.

02

Heterogeneity Analysis

Quantify statistical heterogeneity using Earth Mover's Distance (EMD). Compare IID, label-skew Non-IID, and Dirichlet-based Non-IID partitioning strategies.

03

Convergence & Stability

Measure convergence speed (rounds to 80%), training stability (variance across seeds), and loss behavior over 50 communication rounds.

04

Multi-Dataset Benchmarking

Evaluate on both MNIST (simple, low-dimensional) and CIFAR-10 (complex, RGB images) to test generalizability across task complexities.

05

Statistical Validation

Apply Welch's t-test with Bonferroni correction and Cohen's d effect size analysis to determine whether algorithm differences are statistically significant.

06

Reproducibility

Fix random seeds {42, 52, 62}, implement everything from scratch in PyTorch — no external FL libraries — for full experimental control.

Hyperparameter Configuration

Clients (K)	5
Participation (C)	0.6 $\rightarrow$ 3/5 active per round
Communication Rounds (T)	50
Local Epochs (E)	1
Batch Size	64
Learning Rate	0.01 (SGD)
Momentum	0.9
Weight Decay	1×10^{-4}
Gradient Clip Norm	10.0
Random Seeds	{42, 52, 62}
FedProx μ (sensitivity)	{0.001, 0.01, 0.1}

Datasets & Neural Network Architectures

 MNIST

60,000 train / 10,000 test · 10 classes · 28×28 px
MnistMLP — 235,146 params · 2 FC layers + ReLU + Dropout
Normalization: mean=0.1307, std=0.3081

 CIFAR-10

50,000 train / 10,000 test · 10 classes · 32×32 RGB
CifarCNN — 227,594 params · 3 Conv blocks + BatchNorm
Channel-wise normalization applied

Data Partitioning Strategies

IID

Uniform random split —
balanced classes

EMD = 0.059

Label-Skew Non-IID

2 classes per client —
class exclusion

EMD = 1.938

Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$

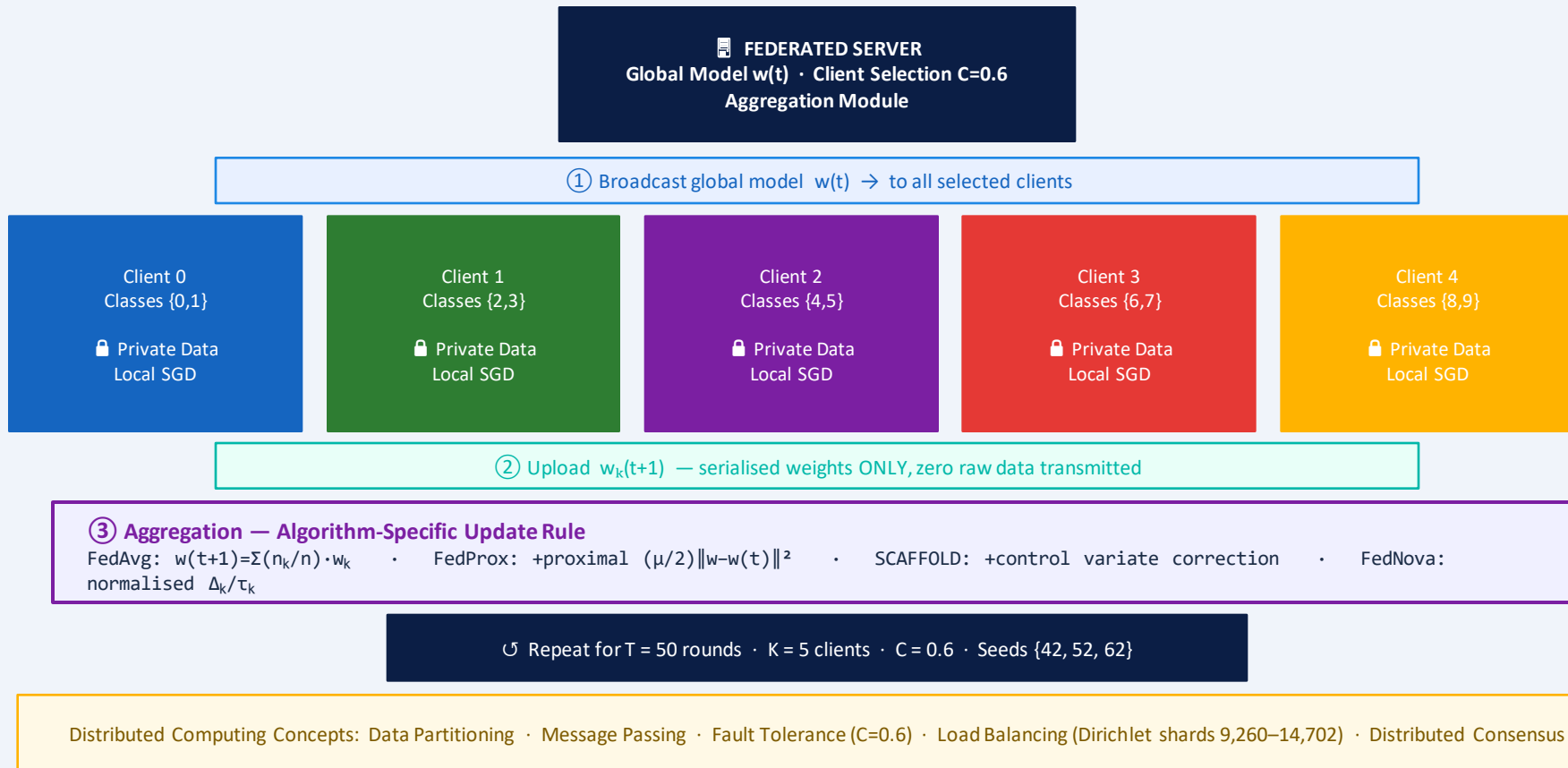
Variable heterogeneity
using Dirichlet dist.

EMD = 0.640



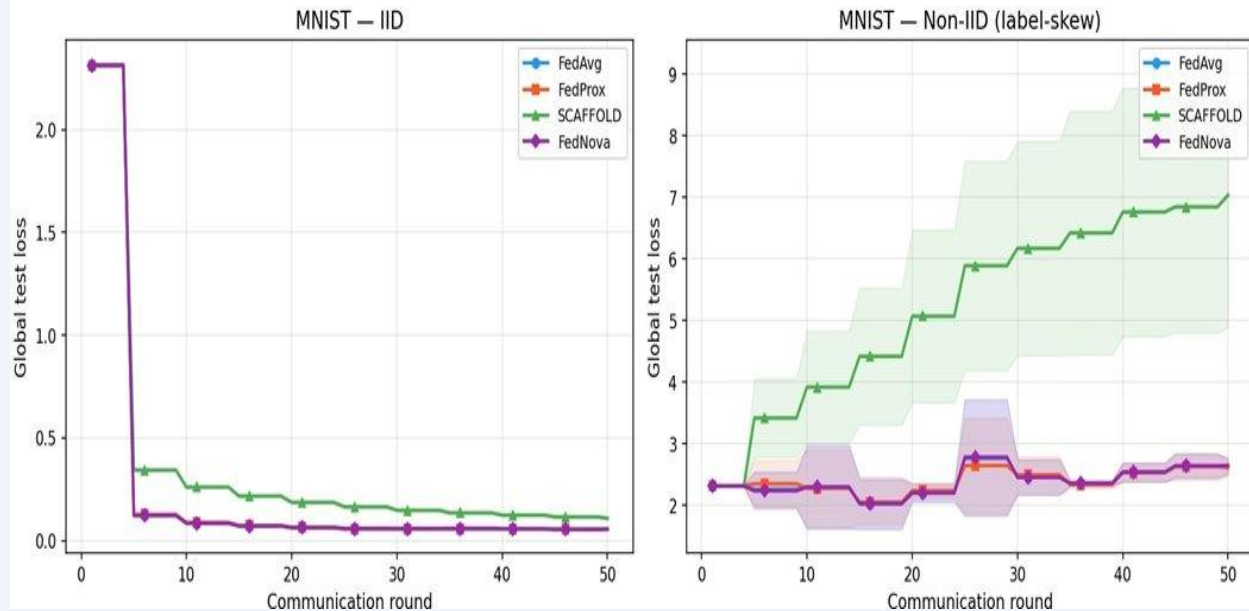
Implemented from scratch in PyTorch 2.8.0 — no external FL libraries

Logical Federated Learning System Architecture



Experiment 1: MNIST Accuracy vs. Communication Rounds

Experiment 1: Loss vs Communication Rounds (MNIST)



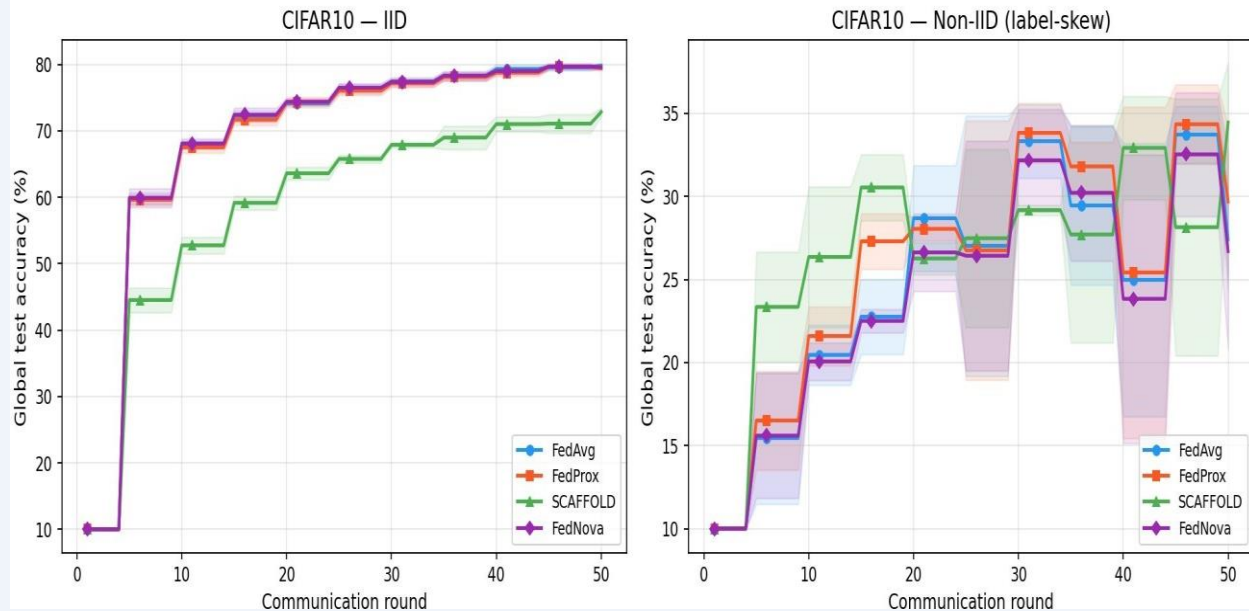
Final Accuracy (Round 50)

IID Results	
FedAvg	98.35%
FedProx	98.33%
FedNova	98.34%
SCAFFOLD	96.64%
Non-IID Results	
FedAvg	65.69%
FedProx	65.73%
FedNova	65.60%
SCAFFOLD	66.02%

✂ **Key Finding:** Under IID, all algorithms reach ~98% in just 5 rounds. Under label-skew Non-IID, all algorithms plateau at 65–66% — and NONE reaches 80% accuracy within 50 rounds. The 32.66 percentage-point drop for FedAvg is caused by the class-exclusion structural barrier, not optimization failure.

Experiment 1: MNIST Loss vs. Communication Rounds

Experiment 1: Accuracy vs Communication Rounds (CIFAR10)



Loss Analysis

IID: Near-zero loss

FedAvg, FedProx, FedNova all converge to loss ≈ 0.056 . SCAFFOLD reaches 0.108 — stable but higher.

Non-IID: Diverging SCAFFOLD

FedAvg, FedProx, FedNova hold loss ≈ 2.6 . SCAFFOLD's loss grows from 3.4 to 7.0 — control variates cannot stabilise under class-exclusion.

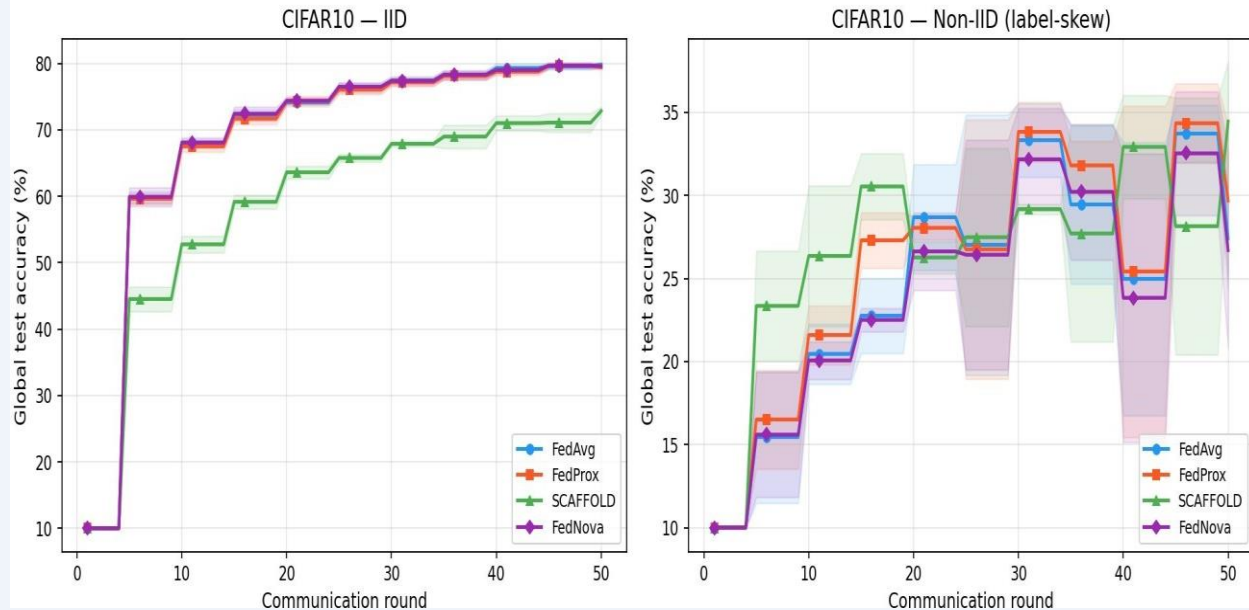
Critical observation

Rising loss with stable accuracy in SCAFFOLD suggests the variate correction is overcorrecting in class-exclusion environments.

Summary: The SCAFFOLD loss divergence under MNIST Non-IID is a critical signal — it reveals that variance correction through control variates is task-dependent. SCAFFOLD is better suited to high-gradient-variance tasks like CIFAR-10 CNNs (confirmed in the next slide).

Experiment 1: CIFAR-10 Accuracy vs. Communication Rounds

Experiment 1: Accuracy vs Communication Rounds (CIFAR10)



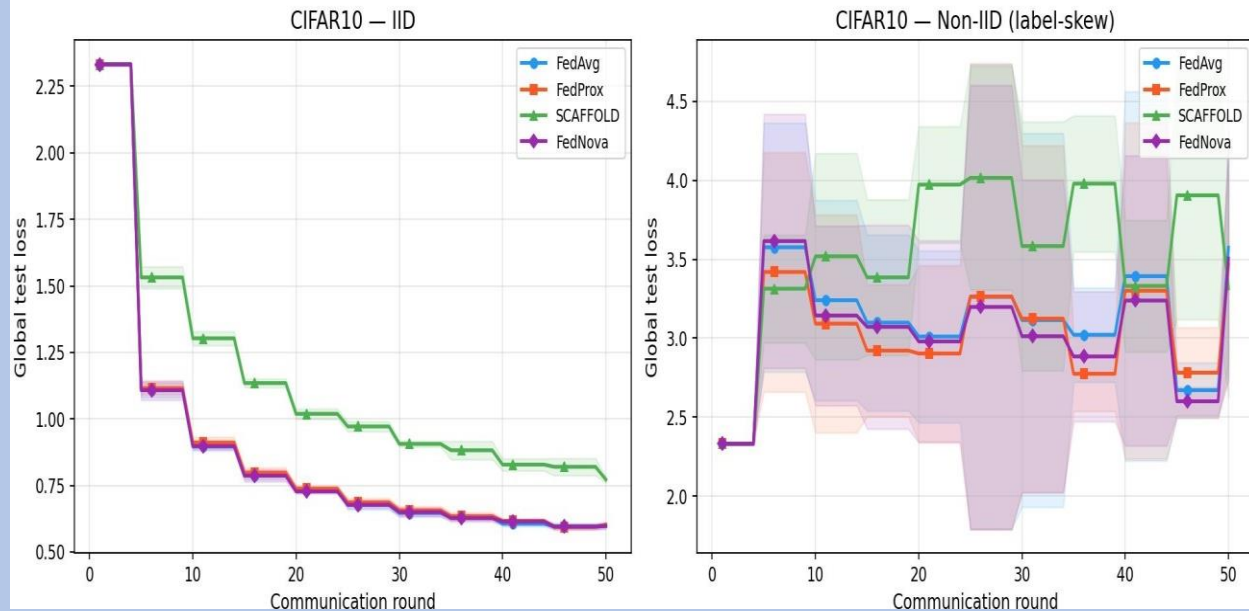
CIFAR-10 Final Accuracy

IID Results	
FedAvg	79.81%
FedProx	79.41%
FedNova	79.58%
SCAFFOLD	72.83%
Non-IID Results	
FedAvg	27.42%
FedProx	29.69%
FedNova	26.71%
SCAFFOLD	34.44% ★

🔄 **Ranking Inversion:** SCAFFOLD is the WORST under CIFAR-10 IID (72.83%) but BEST under Non-IID (34.44%, exceeding FedAvg by 7+ pp). This confirms that variance correction via control variates becomes advantageous in high-gradient-variance CNN tasks under heterogeneous conditions.

Experiment 1: CIFAR-10 Loss vs. Communication Rounds

Experiment 1: Loss vs Communication Rounds (CIFAR10)



Loss Comparison Table

Algorithm	IID Loss	Non-IID Loss
FedAvg	0.598	3.573
FedProx	0.603	3.464
FedNova	0.598	3.504

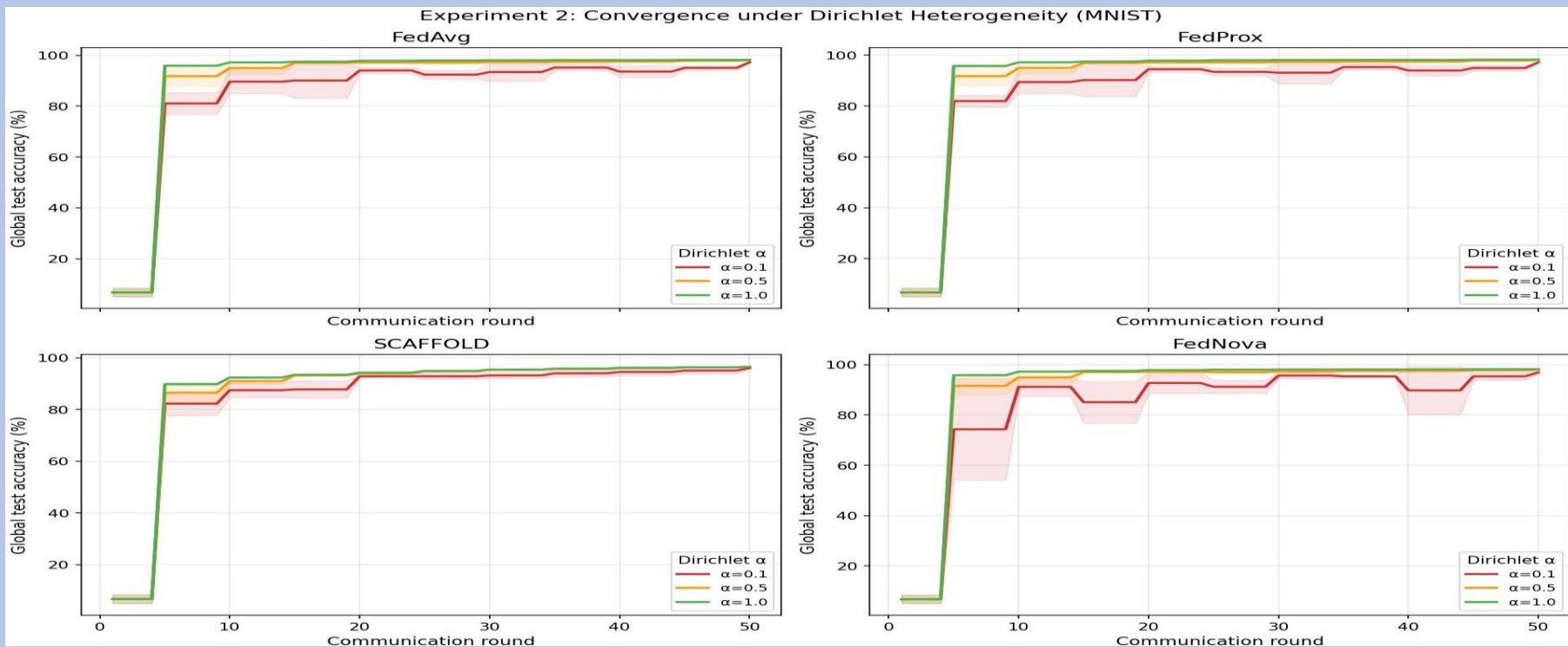
Notable observation:

Under Non-IID, SCAFFOLD achieves the lowest loss (3.315) despite its higher IID loss (0.772). This reinforces the task-specific nature of control variate correction — it excels exactly where gradient variance is highest.

★ **Takeaway:** CIFAR-10 is a significantly harder task than MNIST (complexity from spatial RGB patterns). The algorithms' performance gap under Non-IID widens substantially — from ~0.3 pp spread on MNIST to ~7+ pp spread on CIFAR-10. Dataset complexity amplifies algorithm differences under heterogeneity.

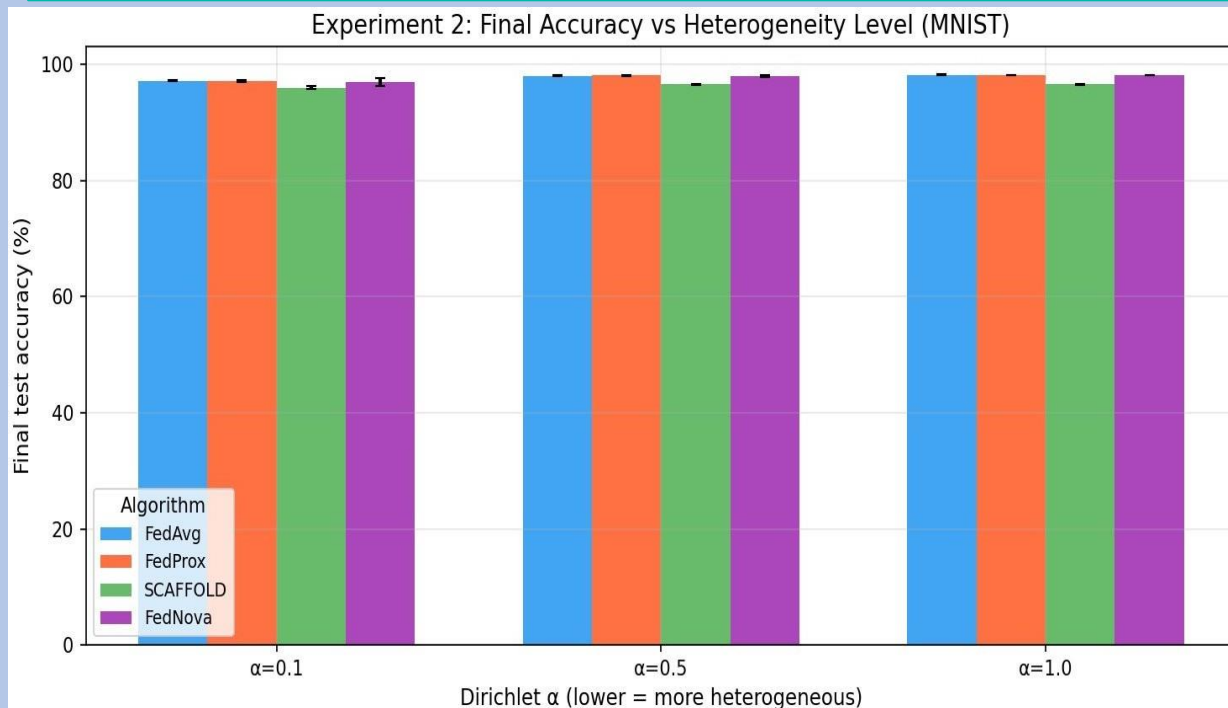
Experiment 2: Convergence under Dirichlet Heterogeneity (MNIST)¹³

$\alpha \in \{0.1, 0.5, 1.0\}$ — lower α = stronger heterogeneity — all four algorithms tested



All four algorithms converge stably under all Dirichlet α values. Even at $\alpha=0.1$ (maximum heterogeneity), convergence is smooth — unlike the chaotic Non-IID label-skew case.

Experiment 2: Final Accuracy vs. Dirichlet Heterogeneity Level



✓ **Conclusion:** Dirichlet partitioning represents a mild form of heterogeneity compared to label-skew. The structural accuracy barrier only emerges when classes are completely absent from clients, not when they are merely underrepresented.

Dirichlet Accuracy Summary

Algorithm	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	Δ
FedAvg	97.25 %	98.08 %	98.22 %	-0. 97
FedProx	97.18 %	98.12 %	98.23 %	-1. 05
FedNova	96.97 %	98.07 %	98.21 %	-1. 24

Tiny degradation

Only 0.54–1.24 pp across the full α sweep. All algorithms maintain >96.9% even at the highest heterogeneity ($\alpha=0.1$).

Dirichlet $\neq$ Label-Skew

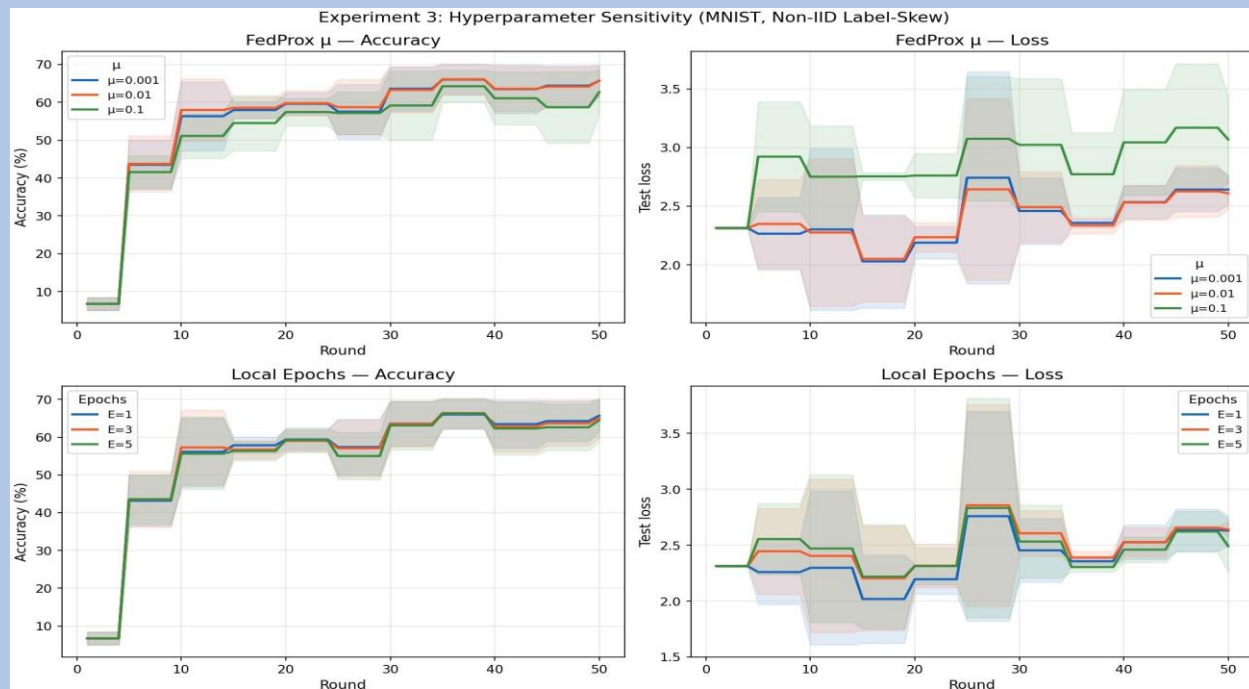
Dirichlet preserves class diversity — every client sees all 10 classes, just in unequal proportions. Label-skew excludes entire classes.

Why the difference?

EMD=0.640 (Dirichlet $\alpha=0.5$) vs EMD=1.938 (label-skew). Class presence alone determines the barrier, not just proportions.

Experiment 3: Hyperparameter Sensitivity Analysis

Testing FedProx proximal coefficient μ and local epochs E — both under Non-IID label-skew on MNIST



μ Sensitivity (FedProx)

μ value	Accuracy
$\mu = 0.001$	$65.70 \pm 4.16\%$
$\mu = 0.01$	$65.73 \pm 3.94\%$
$\mu = 0.1 \triangle$	$62.69 \pm 5.80\% \downarrow$

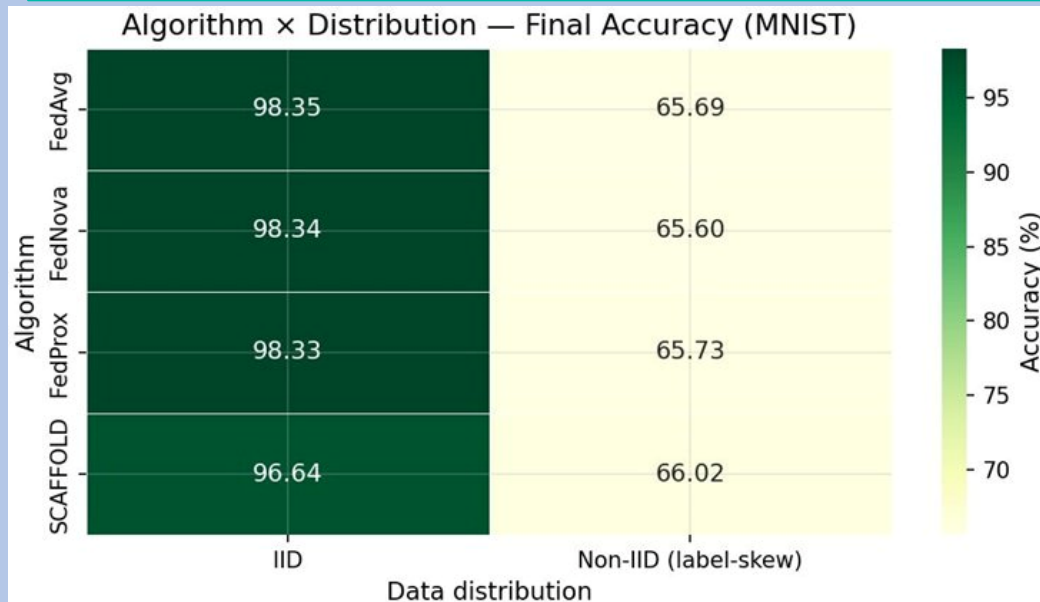
Local Epochs Sensitivity

Epochs (E)	Accuracy
$E = 1$	$65.69 \pm 4.16\%$
$E = 3$	$65.09 \pm 5.08\%$

Both findings confirm that the Non-IID problem is structural. Tuning μ or E cannot break the ~65% barrier — it is caused by class exclusion, not suboptimal hyperparameters.

$\triangle$ Client Drift Confirmed: More local epochs ($E=5$) increase the gap between local and global objectives, worsening client drift. Strong μ (0.1) over-constrains local learning. Both effects are amplified under Non-IID, validating the client drift theory.

Experiment 4: Statistical Validation & Algorithm × Distribution Heatmap



Reading the Heatmap

IID (left column): All algorithms are deep green (96–98%). FL works perfectly.

Non-IID (right column): All algorithms are light yellow (65–66%). They ALL hit the same structural barrier, regardless of the optimization approach used.

Statistical Test Results (MNIST Non-IID)

Alg A	Alg B	p (Bonf.)	Cohen's d
FedAvg	FedProx	1.000	0.011
FedAvg	SCAFFOLD	1.000	0.095
FedAvg	FedNova	1.000	0.020
FedProx	SCAFFOLD	1.000	0.086
FedProx	FedNova	1.000	0.031
SCAFFOLD	FedNova	1.000	0.116

Critical Statistical Finding

All 6 pairwise Bonferroni-corrected p-values = 1.000

All Cohen's d < 0.12 → negligible effect size

No algorithm is statistically or practically better than any other under severe label-skew Non-IID on MNIST.

This project models a realistic distributed federated learning system on a single physical machine using 5 logical clients, demonstrating core distributed computing principles:



Data Partitioning

Training data split into 5 private, disjoint shards — one per client. Each client owns its data exclusively and never shares it. Shard sizes range from 9,260 to 14,702 samples under Dirichlet partitioning.



Message Passing

The only communication allowed is serialized model parameter transmission. No raw data, no gradients — just weight tensors sent as serialized Python objects between the virtual client processes and the server.



Fault Tolerance

Participation fraction $C=0.6$ means 2 out of 5 clients are absent each round, simulating network unavailability, device failures, and communication dropouts in real distributed environments.



Load Balancing

Dirichlet partitioning creates deliberately unequal dataset sizes per client (9,260–14,702 samples). The weighted aggregation $w(t+1) = \sum (n_k/n) \cdot w_k$ compensates for this imbalance automatically.



Distributed Consensus

The aggregation step is equivalent to distributed consensus — all active clients' local model updates are combined into a new global model that no single client can compute alone.



Privacy Preservation

Zero raw data transmitted at any stage. Only serialized model weights are communicated. This satisfies the differential privacy constraint that makes federated learning suitable for regulated industries.

Conclusions & Contributions



IID: All Algorithms Converge Reliably

FedAvg, FedProx, FedNova reach 98.3% on MNIST in just 5 rounds. All algorithms achieve 79–80% on CIFAR-10 IID. Federated optimization is fully effective under homogeneous data.



Structural Non-IID Accuracy Barrier Discovered

All 4 algorithms plateau at 65–66% on MNIST under label-skew. Statistical tests confirm the barrier is not an optimization failure — it is structural, caused by class-exclusion heterogeneity.



Task-Specific SCAFFOLD Behaviour

SCAFFOLD is worst under MNIST IID (96.64%) but BEST under CIFAR-10 Non-IID (34.44%). Variance correction via control variates is more effective when gradient variance is inherently high (CNN tasks).



Dirichlet Heterogeneity Is Fundamentally Different

Only <1.25 pp degradation across the full Dirichlet sweep, versus >32 pp for label-skew. Class presence (not just proportion) is the key factor for the structural barrier.



No Statistically Significant Algorithm Differences

All 6 pairwise Welch t-tests: Bonferroni $p=1.000$, Cohen's $d<0.12$. Under severe heterogeneity, no algorithm meaningfully outperforms another — the data distribution dominates performance.



Reproducible Unified Benchmark

First 4-algorithm comparison with EMD quantification, 3 partitioning strategies, 2 datasets, 3 seeds, and statistical significance — all from scratch in PyTorch 2.8.0.

Future Research Directions & Closing Remarks

01 Larger Federations

Scale to $K > 100$ clients to see if algorithmic differences grow at scale.

02 More Heterogeneity Types

Add covariate shift, feature heterogeneity, temporal distribution drift.

03 Communication Efficiency

Combine gradient compression (QSGD) with FL under Non-IID.

04 CUDA Multi-GPU Validation

Validate on real distributed CUDA infrastructure for wall-clock timing.

05 Personalized FL

FedPer, Ditto — client-specific models to overcome structural barrier.

06 Beyond Classification

NLP, time-series, generative tasks — is the Non-IID barrier universal?

Thank You · Questions & Discussion

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Key References: [1] McMahan et al. FedAvg (AISTATS 2017) · [2] Li et al. FedProx (MLSys 2020) · [3] Karimireddy et al. SCAFFOLD (ICML 2020) · [4] Wang et al. FedNova (NeurIPS 2020) · [5] Hsieh et al. EMD (ICML 2020)